

Hector Contreras

Python Backend Developer | Full-Stack Engineer | QA Tester

San Luis Potosí, México | +52 444 201 7123 | hector12ca.dev@gmail.com | github.com/HectorCntrs | linkedin.com/in/Hector-Contreras-Alvarez

PROFESSIONAL SUMMARY

Intelligent Systems Engineering graduate with hands-on experience in Python backend development, REST API design, full-stack web development, and cloud deployment on AWS. Built and delivered production systems for real clients — including a serverless chatbot on AWS (Lambda, API Gateway, DynamoDB) and a public health platform handling 130M+ medical records. Strong focus on code quality, automated testing, and maintainable solutions. Bilingual (Spanish native, English B2).

TECHNICAL SKILLS

Backend & APIs: Python, FastAPI, C#/ASP.NET, PHP/Laravel, REST APIs, JSON, Webhooks

Frontend & Web: Angular, TypeScript, JavaScript, HTML5, CSS3, Bootstrap

Cloud & Infra: AWS Lambda, API Gateway, DynamoDB, Security Groups, AWS CLF-C02, serverless architecture, Linux/bash

Databases: SQL Server, MySQL, MariaDB — query optimization, large-scale data validation

Testing & QA: xUnit, Moq, Pytest, BDD/Gherkin, white-box & black-box testing, API Testing, Postman

Tools & Practices: Git/GitHub, Jira, Agile/Scrum, JWT, technical documentation

Languages: Spanish (native) · English B2 — Cambridge certified

WORK EXPERIENCE

KaraOkulta – An Amber Studio — Game QA & Developer Intern

Guadalajara, México (Remote) | Jun 2025 – Dec 2025

- Developed modular software systems in Lua using OOP principles, reusable components, and Git version control across 3+ active production game systems.
- Performed structured manual testing on PC and mobile platforms to identify root causes of UI, logic, and performance incidents; delivered written diagnostic reports with findings and resolution paths.
- Coordinated with stakeholders to clarify expected vs. actual system behavior, align on incident priority, and confirm acceptance criteria before closing tickets.
- Collaborated in Agile/Scrum sprints to coordinate feature delivery, issue triage, and cross-functional blockers across a 6-month engagement.

LaNGIF – National Geoprocessing Laboratory — Full-Stack Developer | QA Tester

San Luis Potosí, México | Aug 2025 – May 2026 | Project-based contract

- Developed the medical records CRUD module end-to-end — data listing view, validated entry and edit forms — covering both Angular frontend and ASP.NET backend layers.
- Wrote white-box unit tests with xUnit and Moq covering service logic, repository methods, DTO mapping, and error handling; performed black-box testing on the frontend validating forms, filters, pagination, and UI consistency.
- Optimized SQL Server queries to support paginated, filterable access to a 130M+ record epidemiological dataset, enabling lookups by patient ID, region, and year without performance degradation.
- Implemented Python scripts to detect and correct null values in required fields across the dataset prior to database ingestion, contributing to the team's data quality pipeline.
- Collaborated in Agile/Scrum sprints with a 6-person team using Jira for ticket tracking, issue documentation, and sprint planning.

PROJECTS

WhatsApp Automation Chatbot — Roberto Stickers

Feb 2026 – Apr 2026 | Python, AWS Lambda, API Gateway, DynamoDB, WhatsApp Cloud API, BDD/Gherkin

- Developed and deployed a serverless WhatsApp chatbot for a real sticker business client on AWS — handling predefined response flows for frequent customer questions about products, prices, and shipping.
- Implemented detection logic for unanswered questions and an automatic owner notification flow delivering the customer's phone number and full message context for human follow-up.
- Designed BDD/Gherkin test scenarios to validate conversation flows, fallback behavior, and notification triggers across happy paths and edge cases.

UASLP — Social Service: Fake News Detection System

Feb 2025 – Aug 2025 | Python, HTML, CSS, JavaScript

- Validated a FastAPI REST service for real-time text classification — testing request handling, response structure, prediction output, and error scenarios across 10+ documented test cases.
- Identified and escalated model output inconsistencies (false positives, preprocessing failures, API mismatches) with full reproduction steps; generated validation reports documenting classification accuracy and edge-case behavior.

RefuPet — Pet Adoption Web Platform

Apr 2026 – May 2026 | PHP/Laravel, MySQL, REST API, Bootstrap, Postman

- Built a full-stack pet adoption platform with 3 CRUD modules (animals, adoption requests, users), server-side validation with Laravel Form Requests, and paginated list views.
- Implemented role-based access control (admin vs. adopter) with Spatie Permissions, REST API with Laravel Sanctum, dynamic PDF reports with DomPDF, and automated confirmation emails with Laravel Queues.

Swimming Event Registration System — Consejo Supremo Indígena de Michoacán

Apr 2026 | Angular, TypeScript, Google Apps Script, Google Sheets

- Delivered a responsive Angular/TypeScript registration frontend for a real public swimming event, with a validated form and automated confirmation emails with unique folio numbers per participant.

- Integrated Google Apps Script and Google Sheets as a zero-cost serverless backend for participant data storage and processing.

EDUCATION & CERTIFICATIONS

Universidad Autónoma de San Luis Potosí — Intelligent Systems Engineering

San Luis Potosí, México | Aug 2021 – May 2026 | Degree completed

AWS Certified Cloud Practitioner (CLF-C02) — Amazon Web Services

Jul 2026

CCNAv7: Introduction to Networks (Jun 2024) · CCNA Switching, Routing & Wireless Essentials (Jul 2025) — Cisco / UASLP